Alice

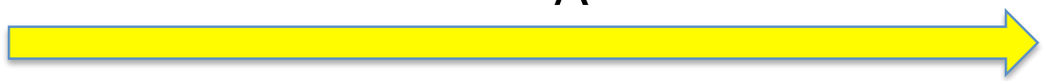
Bob

S (TTP)

k_{AS}

k_{BS}

k_{AS}, k_{BS}

new N_A

A, N_A